Q1.

(a) The chemical formula for hydrochloric acid is HCl.
The chemical formula for sodium hydroxide is NaOH.

When they react together, two products are formed.
The chemical formula for one product is NaCl.

(i) Complete the word equation below with the **names** of both products.

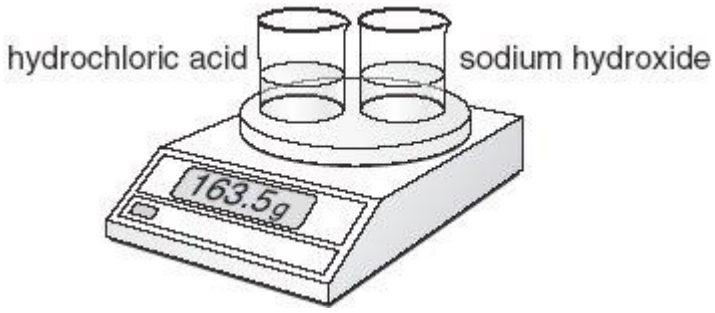
1 mark

(ii) **On the dotted line**, give the chemical formula of the other product.

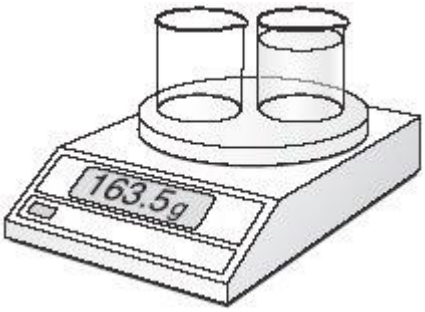
sodium hydroxide	+	hydrochloric acid	⟶	_____	+	_____
NaOH		HCl		NaCl		.....

2 marks

(b) In experiment 1, Molly put two beakers on a balance.
One contained 20 cm^3 of hydrochloric acid.
The other contained 20 cm^3 of sodium hydroxide solution.
The total mass was 163.5 g.



She poured the acid onto the sodium hydroxide. They reacted.



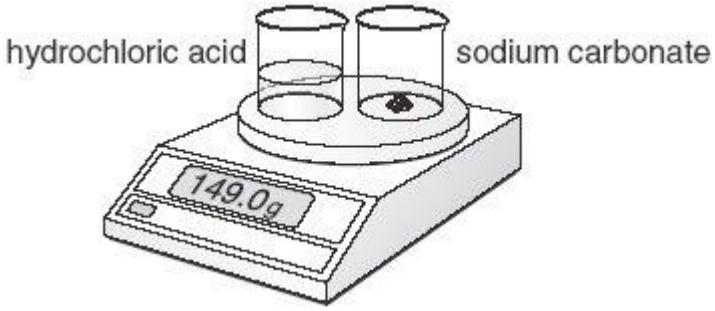
Why did the reading on the balance **not** change?

.....

.....

1 mark

(c) In experiment 2, Molly put two beakers on a balance.
One contained 20 cm^3 of hydrochloric acid.
The other contained 5 g of sodium carbonate.



She poured the acid onto the sodium carbonate. They reacted.
Two of the products are the same as in experiment 1.

(i) Complete the word equation with the names of the **three** products.

sodium carbonate + hydrochloric acid → ..... + ... + ...

1 mark

(ii) The total mass at the start was 149.0 g.
When the reaction stopped, the reading on the balance was 147.0 g.

Why was there a loss of mass in this reaction?

.....

.....

1 mark
maximum 6 marks

Q1.

(a) (i) • sodium chloride + • water

accept 'hydrogen oxide' for water

answers must be in the correct order

2 (L7)

(ii) • H_2O

1 (L7)

(b) any **one** from

• the same atoms are present in the reactants and the products

accept 'mass is conserved'

'the mass did not change' is insufficient
'the particles are the same' is insufficient
*do **not** accept 'the molecules are the same'*

• no gas was given off

*accept 'nothing was added **or** lost'*

1 (L7)

(c) (i) • sodium chloride
carbon dioxide
water

answers may be in any order
***all three** answers are required for the mark*

1 (L7)

(ii) any **one** from

• carbon dioxide **or** gas escapes

'mass is lost' is insufficient as it is given in the question

• carbon dioxide is a gas

• one of the products is a gas **or** escapes

1 (L7)

[6]